

INCIDENT RESPONSE REPORT

Simulating and Responding to a Multi-Stage APT Attack
Using Open-Source SOC Tools

Project Type	Internship Final Project
Primary Detection Platform	Wazuh EDR / SIEM
Laboratory Environment	Kali Linux attacker and Windows 10 victim
Report Date	July 2026

Executive Summary

This report documents the simulation, detection, investigation, and response to a multi-stage Advanced Persistent Threat (APT) attack conducted in a controlled laboratory environment using open-source Security Operations Center (SOC) technologies. The objective of the project was to understand the complete attack lifecycle, evaluate endpoint detection capabilities, and demonstrate incident response procedures using Wazuh as the primary Endpoint Detection and Response (EDR) and Security Information and Event Management (SIEM) solution.

The simulated attack originated from a Kali Linux attacker machine targeting a Windows 10 endpoint through a phishing-based payload delivery. After gaining initial access, the attacker established a Meterpreter reverse shell, performed system reconnaissance, created persistence through the Windows Startup folder, and simulated data exfiltration over HTTP.

Wazuh successfully detected multiple stages of the attack by generating alerts related to malicious process creation, PowerShell execution, scheduled task activity, and suspicious network communications. Custom YARA and Sigma detection rules were also developed to improve detection accuracy.

Although additional SOC platforms such as TheHive, Cortex, MISP, and Security Onion could not be fully deployed due to infrastructure and compatibility issues, Wazuh provided comprehensive endpoint visibility, enabling successful incident detection and investigation.

1. Incident Overview

1.1 Incident Description

A controlled multi-stage APT attack simulation was conducted to evaluate the effectiveness of open-source SOC technologies in detecting and responding to advanced cyber threats. The attack followed a realistic intrusion lifecycle beginning with phishing-based payload delivery and progressing through execution, persistence, discovery, and simulated data exfiltration.

The attack was performed entirely within a virtual laboratory environment to avoid impact on production systems.

1.2 Environment Details

Component	Details
Attacker Machine	Kali Linux
Attacker IP	192.168.56.102
Victim Machine	Windows 10
Victim IP	192.168.56.101
Detection Platform	Wazuh
Visualization Platform	Kibana (Wazuh Dashboard)
Attack Vector	Phishing simulation using malicious payload
Persistence Method	Windows Startup Folder

1.3 Scope of Impact

The simulated compromise affected the Windows 10 endpoint by:

- Executing unauthorized code

- Establishing remote command execution
- Performing system reconnaissance
- Creating persistence mechanisms
- Simulating outbound data exfiltration

No production assets or external systems were involved.

2. Attack Timeline

Time	Attack Stage	Description
T0	Reconnaissance	Attacker generated a malicious executable (payload.exe) using msfvenom on Kali Linux.
T1	Initial Access	Victim downloaded the payload over HTTP from the attacker's machine.
T2	Execution	Victim executed payload.exe, establishing a Meterpreter reverse shell back to the attacker.
T3	Discovery	Attacker collected system information using Meterpreter commands such as sysinfo and getuid.
T4	Persistence	The payload was copied into the Windows Startup folder to ensure execution upon user login.
T5	Exfiltration	Simulated data exfiltration was performed over HTTP to the attacker system.
T6	Detection	Wazuh generated multiple alerts identifying suspicious activities throughout the attack chain.

3. Indicators of Compromise (IoCs)

IoC Type	Value
Malicious File	payload.exe
MD5	6745e064109110c05abf6b45d1fd0591
SHA256	5a939855519072752fd9168237125880e85aa64d743641ab0ee24bfae96d542f
Attacker IP	192.168.56.102
Victim IP	192.168.56.101
Persistence Location	C:\Users\vania\AppData\Roaming\Microsoft\Windows\Start Menu\Programs\Startup\payload.exe
Communication Protocol	HTTP
Attack Tool	Meterpreter
Payload Generator	msfvenom

4. Detection & Investigation

4.1 Detection Method

Wazuh served as the primary detection platform and continuously monitored endpoint activities on the Windows 10 system. During the simulated attack, Wazuh successfully correlated multiple security events and generated alerts corresponding to different stages of the attack.

4.2 Observed Security Events

1. Malicious Process Creation

Wazuh detected the execution of the suspicious executable: payload.exe. This indicated successful payload execution on the victim endpoint.

2. PowerShell Activity

Suspicious execution of powershell.exe was identified, indicating possible post-exploitation activity commonly associated with attacker automation and lateral movement.

3. Scheduled Task Detection

Wazuh detected scheduled task creation (schtasks) that could be used for persistence or automated execution of malicious code.

4. Network Connection Monitoring

Suspicious outbound communications were detected between the victim 192.168.56.101 and attacker 192.168.56.102. These network events aligned with the establishment of the Meterpreter reverse shell and simulated data exfiltration.

4.3 Custom Detection Rules

YARA Rule

```
rule Detect_Payload {
  meta:
    description = "Detects payload.exe used in APT simulation"
    author = "Intern"
    date = "2026-07-30"

  strings:
    $name = "payload.exe" ascii
    $path = "Startup\\payload.exe" ascii

  condition:
    $name or $path
}
```

Purpose

- Detects the presence of the malicious executable.
- Detects persistence through the Windows Startup folder.

Sigma Rule

```
title: Suspicious Scheduled Task Creation
status: experimental

description: Detects creation of scheduled tasks with suspicious names

logsource:
  product: windows
  service: security

detection:
  selection:
    EventID: 4698
    TaskContent|contains:
      - 'payload.exe'
      - 'powershell'

  condition: selection
```

Purpose

- Detects malicious scheduled task creation.
- Identifies persistence mechanisms involving PowerShell or the payload.

4.4 Investigation Findings

Analysis of endpoint logs confirmed that the attacker successfully completed the following stages:

- Initial compromise
- Payload execution
- Remote shell establishment
- Host reconnaissance
- Persistence creation
- Simulated data exfiltration

The generated Wazuh alerts accurately reflected these activities, enabling analysts to reconstruct the complete attack timeline.

No evidence of privilege escalation or lateral movement was introduced as part of this simulation.

5. Tool Deployment Notes

5.1 Successfully Deployed: Wazuh

Status: Successful

Wazuh was successfully installed and configured as the central security monitoring platform.

Capabilities demonstrated:

- Endpoint monitoring
- Process creation detection
- Network activity monitoring
- Security event correlation
- Alert generation
- Log visualization through Kibana
- Custom rule integration (YARA and Sigma)

Wazuh served as the primary tool for incident detection and investigation throughout the project.

5.2 Deployment Challenges

Platform	Status	Notes
TheHive	Unsuccessful	Multiple deployment attempts were made using Docker containers and native .deb packages. SSL/TLS authentication failures occurred between TheHive and the Wazuh Elasticsearch Indexer. Despite extensive troubleshooting, a stable and authenticated connection could not be established.
Security Onion	Unsuccessful	Deployment could not be completed because the ISO download repeatedly failed due to unstable network connectivity, and downloaded images became corrupted during transfer. Installation was not possible within the project timeline.
Cortex	Not Deployed	Cortex depends on a functioning TheHive deployment. Since TheHive installation was unsuccessful, Cortex integration could not proceed.
MISP	Not Deployed	MISP deployment was planned for threat intelligence integration but was postponed due to the inability to deploy TheHive and Cortex.

6. Mitigation Recommendations

1. Enable PowerShell Logging

Enable Script Block Logging and Module Logging to improve visibility into malicious PowerShell activity.

2. Monitor Windows Startup Folder

Continuously monitor Startup directories for unauthorized executables or modifications that may indicate persistence attempts.

3. Block Malicious IP Addresses

Configure firewall rules to block communication with known malicious or unauthorized external IP addresses.

4. Enable Process Creation Logging

Enable Windows Event ID 4688 to capture detailed process creation events for endpoint monitoring and forensic investigations.

5. Implement Application Whitelisting

Restrict execution of unauthorized binaries using technologies such as Windows Defender Application Control (WDAC) or AppLocker.

6. Regular Patch Management

Maintain current operating system and application patches to reduce the attack surface and mitigate known vulnerabilities.

7. Deploy Multi-Layer Monitoring

Integrate endpoint detection, network monitoring, and centralized log analysis to improve detection coverage across all stages of the attack lifecycle.

8. Strengthen Email Security

Implement phishing protection mechanisms, attachment scanning, and user awareness training to reduce the likelihood of successful phishing attacks.

7. Conclusion

This project successfully demonstrated the complete lifecycle of a simulated multi-stage Advanced Persistent Threat (APT) attack within a controlled laboratory environment. Beginning with phishing-based payload delivery and progressing through execution, persistence, reconnaissance, and simulated data exfiltration, the attack mirrored several techniques commonly observed in real-world cyber intrusions.

Wazuh proved to be an effective open-source EDR and SIEM platform by providing comprehensive endpoint visibility, generating timely alerts, and enabling detailed log analysis through the Kibana dashboard. The development of custom YARA and Sigma rules further enhanced detection capabilities by identifying malicious payloads and suspicious scheduled task creation.

Although the deployment of TheHive, Security Onion, Cortex, and MISP could not be completed due to technical and infrastructure-related challenges, these limitations did not prevent the successful execution of the core objectives. The project demonstrated practical skills in attack simulation, log analysis, indicator identification, detection engineering, and incident response.

Overall, the exercise highlights the importance of layered security monitoring, proactive detection mechanisms, and continuous improvement of defensive controls in identifying and responding to sophisticated cyber threats. The experience gained through this project provides a strong foundation for real-world SOC operations, digital forensics, and incident response activities.